

An Exact Dimension Threshold for the Localized Aubin–Talenti Family in arXiv:2504.15968

Candidate full solution produced in run `fa_banach_001`
Model: GPT5.6

19 July 2026

Status. Candidate full solution, currently assessed as likely valid and awaiting human review. The novelty search described below was bounded, not exhaustive.

1 Source and open question

The source is S. Chakraborty, D. Gupta, S. Malhotra, and K. Sreenadh, *Global Compactness Result for a Brézis–Nirenberg-Type Problem Involving Mixed Local Nonlocal Operator*, arXiv:2504.15968v2 (11 May 2025) [1]. On page 7, Remark 1.1(1), the authors ask for the threshold dimension beyond which the energy of their localized Aubin–Talenti family (1.7) is strictly below $2^{2/N}\mathcal{S}_N$.

it allows us to deform the bubble family so that its energy approaches the first critical level. Assuming by contradiction that no positive nontrivial solution exists, this level $2^{2/N}\mathcal{S}_N$ becomes the next critical threshold after $\mathcal{S}_N$ at which the Palais–Smale condition breaks down. Thus, applying a version of the deformation lemma, we construct a homotopy retracting the unit sphere $\mathbb{S}^{N-1} \subset \Omega$ continuously to a single point within Ω . However, such a deformation contradicts the topological condition (1.6).

Remark 1.1. (1) *One of the natural limitations of our result lies in the assumption that the dimension N must be sufficiently large in order to guarantee the existence of solutions. While this assumption is crucial for our analytical estimates to hold, numerical simulations performed using MATLAB suggest that Lemma 4.3 may fail for lower dimensions. This raises an intriguing open question: to precisely determine the threshold dimension beyond which the energy associated with the localized family of Aubin–Talenti bubbles defined in (1.7) falls strictly below the critical level $2^{2/N}\mathcal{S}_N$.*

(2) *An alternative approach to relax the large dimension assumption is to introduce a small parameter $\varepsilon = \varepsilon(N)$ in front of the fractional Laplacian operator. By tuning ε appropriately, one can ensure that the energy lies below the second*

Figure 1: The open question in arXiv:2504.15968v2, page 7, Remark 1.1(1). The crop retains the full readable page width.

In the paper’s notation, the unnormalized Gagliardo seminorm is

$$[u]_s^2 = \iint_{\mathbb{R}^{2N}} \frac{|u(x) - u(y)|^2}{|x - y|^{N+2s}} \, dx \, dy, \quad 0 < s < 1,$$

and the mixed Rayleigh quotient is

$$S(u; \mathbb{R}^N) = \frac{\|\nabla u\|_2^2 + [u]_s^2}{\|u\|_{2^*}^2}, \quad 2^* = \frac{2N}{N-2}.$$

The standard bubble is

$$U_N(x) = (1 + |x|^2)^{-(N-2)/2}.$$

2 Proof intuition

The source estimates $[U_N]_s$ by interpolating between the L^2 and gradient norms. That is sufficient asymptotically but loses the dimension at which the inequality first becomes true. Here the seminorm is evaluated exactly. The unitary Fourier transform of U_N is a multiple of $|\xi|^{-1}K_1(|\xi|)$, so the fractional energy is a Mellin integral of K_1^2 and hence a product of Gamma functions. After division by the exact gradient energy, the source question becomes one explicit scalar inequality.

The remaining issue is whether success can later revert to failure as N grows. The quotient between dimensions $N+2$ and N collapses to an elementary rational expression. Once normalized by the critical barrier $2^{2/N} - 1$, this quotient is strictly less than one for $N \geq 7$. Thus the two parity classes are monotone: two consecutive successful dimensions certify every larger dimension and give a finite exact stopping rule.

3 Exact theorem

For $N \geq 5$ and $0 < s < 1$, set

$$\begin{aligned} Q_{N,s} &:= \frac{[U_N]_s^2}{\|\nabla U_N\|_2^2} \\ &= \frac{2\pi^{N/2}|\Gamma(-s)|\Gamma(\frac{N}{2} + s - 1)^2\Gamma(\frac{N}{2} + s - 2)\Gamma(N)}{N(N-2)\Gamma(\frac{N}{2} - 1)^2\Gamma(\frac{N}{2})^2\Gamma(N+2s-2)}, \end{aligned} \quad (1)$$

and

$$h_N := 2^{2/N} - 1, \quad F_{N,s} := \frac{Q_{N,s}}{h_N}.$$

Let $P_N(s)$ denote the precise uniform large-cutoff property used in the source's Lemma 4.3: there are $S_1 < 2^{2/N}\mathcal{S}_N$ and R_0 such that the normalized localized family (1.7) has Rayleigh quotient at most S_1 whenever $R \geq R_0$, uniformly in $t \in [0, 1)$ and $\sigma \in \mathbb{S}^{N-1}$.

Theorem 1 (Exact threshold and finite stopping rule). *For each $s \in (0, 1)$ the following statements hold.*

1. For every $N \geq 5$,

$$P_N(s) \iff Q_{N,s} < 2^{2/N} - 1 \iff F_{N,s} < 1.$$

2. For every $N \geq 7$,

$$F_{N+2,s} < F_{N,s}.$$

Consequently both parity subsequences are strictly decreasing from dimension 7 onward and tend to zero.

3. The exact eventual threshold is

$$N_{\text{th}}(s) = 1 + \max\{N \geq 5 : F_{N,s} \geq 1\}.$$

The set in (3) is finite. It is determined by a finite procedure: evaluate the explicit formula (1) in increasing dimension, retaining the earlier failures, and stop after two consecutive successes beginning at a dimension at least 7. No later failure is possible, so one plus the last retained failure is exactly $N_{\text{th}}(s)$.

Thus the question is reduced without estimates to a closed Gamma-function test, together with a proved finite stopping rule. For illustration, the included 80-digit checker gives the following values.

s	0.05	0.10	0.25	0.50	0.75	0.90	0.99
$N_{\text{th}}(s)$	22	21	19	19	20	22	25

These decimal evaluations are checks of the exact criterion, not ingredients of the proof.

4 Proof of Theorem 1

4.1 Exact fractional energy

Use the unitary Fourier transform

$$\widehat{f}(\xi) = (2\pi)^{-N/2} \int_{\mathbb{R}^N} e^{-ix \cdot \xi} f(x) \, dx.$$

Write $B = N/2$ and $A = B + s - 1$. The Gamma integral representation of $(1 + |x|^2)^{-(B-1)}$, followed by the Gaussian Fourier transform, gives

$$\widehat{U}_N(\xi) = \frac{2^{2-B}}{\Gamma(B-1)} \frac{K_1(|\xi|)}{|\xi|}. \quad (2)$$

Indeed the remaining Laplace integral is

$$\int_0^\infty t^{-2} e^{-t-|\xi|^2/(4t)} \, dt = \frac{4K_1(|\xi|)}{|\xi|}.$$

Let

$$c_{N,s} = \frac{4^s \Gamma(B+s)}{\pi^B |\Gamma(-s)|}.$$

This is the multiplier normalization for the fractional Laplacian, and Plancherel gives the raw-seminorm identity

$$[f]_s^2 = \frac{2}{c_{N,s}} \int_{\mathbb{R}^N} |\xi|^{2s} |\widehat{f}(\xi)|^2 \, d\xi. \quad (3)$$

The standard Mellin integral [2, Section 10.43], valid for $\Re \mu > 2|\Re \nu|$, is

$$\int_0^\infty r^{\mu-1} K_\nu(r)^2 \, dr = \frac{2^{\mu-3}}{\Gamma(\mu)} \Gamma\left(\frac{\mu}{2}\right)^2 \Gamma\left(\frac{\mu}{2} + \nu\right) \Gamma\left(\frac{\mu}{2} - \nu\right). \quad (4)$$

Apply (4) with $\mu = N + 2s - 2 = 2A$ and $\nu = 1$. Equations (2)–(3), together with $\omega_{N-1} = 2\pi^B/\Gamma(B)$, yield

$$[U_N]_s^2 = \frac{\omega_{N-1} \pi^B |\Gamma(-s)| \Gamma(A)^2 \Gamma(A-1)}{\Gamma(B-1)^2 \Gamma(2A)}. \quad (5)$$

All conditions are satisfied for $N \geq 5$ and $0 < s < 1$.

A radial beta integral gives

$$\|\nabla U_N\|_2^2 = \frac{\omega_{N-1} N(N-2)}{2} \frac{\Gamma(B)^2}{\Gamma(N)}. \quad (6)$$

Dividing (5) by (6) proves (1).

4.2 Equivalence with the localized family

The source proves uniform convergence of its cut-off bubbles to the uncut translated/dilated family in the gradient norm, the Gagliardo seminorm, and the critical norm [1, Section 4]. Translation and critical scaling give

$$\begin{aligned}\|\nabla U[t\sigma, 1-t]\|_2^2 &= \|\nabla U_N\|_2^2, \\ \|U[t\sigma, 1-t]\|_{2^*}^2 &= \|U_N\|_{2^*}^2, \\ [U[t\sigma, 1-t]]_s^2 &= (1-t)^{2-2s}[U_N]_s^2.\end{aligned}$$

Since U_N attains the sharp classical Sobolev quotient,

$$S(U[t\sigma, 1-t]; \mathbb{R}^N) = \mathcal{S}_N(1 + (1-t)^{2-2s}Q_{N,s}).$$

The supremum over $t \in [0, 1)$ occurs at $t = 0$. Therefore its limiting value is below $2^{2/N}\mathcal{S}_N$ exactly when $Q_{N,s} < h_N$. A strict inequality leaves a positive gap, and uniform cutoff convergence supplies the required S_1 and R_0 . Conversely, if $Q_{N,s} \geq h_N$, the $t = 0$ cutoffs converge to a value at or above the threshold, so no fixed $S_1 < 2^{2/N}\mathcal{S}_N$ can work for all large R . This proves part 1.

4.3 Monotonicity in dimension

Gamma recurrence in (1) simplifies completely:

$$\frac{Q_{N+2,s}}{Q_{N,s}} = \frac{2\pi(N+1)(N+2s-2)(N+2s-4)}{(N+2s-1)N(N-2)(N+2)}. \quad (7)$$

For $N \geq 7$, the factor $x(x-2)/(x+1)$ is increasing at $x = N+2s-2$, hence the right side of (7) is strictly below its $s \uparrow 1$ limit:

$$\frac{Q_{N+2,s}}{Q_{N,s}} < \frac{2\pi}{N+2}. \quad (8)$$

It remains to compare the barriers. Put $h_N = 2^{2/N} - 1$. For $N \geq 7$, $h_N \leq h_7 < 11/50$. The latter inequality follows by raising $2^{2/7} < 61/50$ to the seventh power, since $(61/50)^7 > 4$. The classical rational bounds $\log 2 > 173/250$ and $\pi < 22/7$ imply $\log 2/\pi > 11/50$. Consequently

$$2\pi h_N < 2\log 2 < (N+2)h_{N+2},$$

where the second inequality uses $e^x - 1 > x$ at $x = 2\log 2/(N+2)$. Combining this with (8) gives

$$\frac{F_{N+2,s}}{F_{N,s}} = \frac{Q_{N+2,s}}{Q_{N,s}} \frac{h_N}{h_{N+2}} < 1.$$

For completeness, $e^x - 1 < xe^x$ and $e^y - 1 > y$ also give

$$\frac{h_N}{h_{N+2}} < \frac{N+2}{N} 2^{2/N} < 2 \quad (N \geq 7).$$

Together with (8), this makes the quotient $F_{N+2,s}/F_{N,s}$ of order $O(1/N)$, so both parity subsequences tend to zero. This proves part 2, finiteness in (3), and the stopping rule in part 3.

5 Verification notes

The included script `code/threshold_verifier.py` evaluates (1) in log-Gamma form, verifies (7) against independent Gamma evaluations, and checks representative thresholds. With 80 decimal digits, the maximum relative recurrence discrepancy over the sampled cases was below 2×10^{-78} . Direct numerical quadrature of the Bessel integral at $(N, s) = (5, 1/2), (8, 1/4), (19, 3/4)$ also agreed with (1).

There is a useful normalization check independent of those calculations. Taking $s \uparrow 1$ in (5) gives

$$\lim_{s \uparrow 1} (1-s) \frac{[U_N]_s^2}{\|\nabla U_N\|_2^2} = \frac{\omega_{N-1}}{2N},$$

which is exactly the Bourgain–Brezis–Mironescu constant for the raw double integral used by the source.

6 Novelty search, scope, and limitations

A bounded search was performed on 19 July 2026. It covered the exact arXiv id 2504.15968, the quoted phrase “precisely determine the threshold dimension”, “localized Aubin–Talenti bubbles”, “mixed local nonlocal”, and the level $2^{2/N} \mathcal{S}_N$, using arXiv-focused and general web search. The current source remains arXiv v2, last revised 11 May 2025 [1]. The search found related later work on mixed local–nonlocal critical problems but no paper claiming this exact threshold calculation. Thus no known duplicate was found, but the search is not an exhaustive bibliographic guarantee.

This result determines the threshold for the specific localized family and the uniform fixed-gap statement used in the source. It does not claim that a different test family cannot establish the PDE existence theorem in lower dimensions, nor does it relax the domain assumptions in Remark 1.1(3).

7 Human-review recommendation

The result is suitable for expert review as a candidate full answer to Remark 1.1(1). The highest-value checks are: (i) the factor $2/c_{N,s}$ in (3); (ii) the cancellation leading to (5); and (iii) that the source’s uniform cutoff lemma is interpreted with a fixed $S_1 < 2^{2/N} \mathcal{S}_N$, so equality at the uncut limit is correctly classified as failure.

References

- [1] S. Chakraborty, D. Gupta, S. Malhotra, and K. Sreenadh, *Global Compactness Result for a Brézis–Nirenberg-Type Problem Involving Mixed Local Nonlocal Operator*, arXiv:2504.15968v2, 2025.
- [2] NIST Digital Library of Mathematical Functions, *Modified Bessel Functions: Integrals*, Section 10.43, <https://dlmf.nist.gov/10.43>, accessed 19 July 2026.